GOVERNMENT OF GUJARAT · INFORMATION TECHNOLOGY DEPARTMENT

Gujarat Government Centralized Video Intelligence Platform

High-Level Design (HLD) — System Architecture Document

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1. PROJECT OBJECTIVE

A centralized platform that integrates existing government camera infrastructure across Gujarat through a secure middleware layer, providing centralized monitoring, GIS-based camera visualization and AI-powered video intelligence for authorized government departments.

2. SCOPE & TARGET COVERAGE

Dimension	Platform Specification
Participating Departments	GSRTC, Police, RTO, Hospital, and other designated state administrative departments
Camera Capacity	Starts at baseline (~60 cameras) architected to scale toward 80,000 cameras via incremental benchmarking
Geographic Scope	Statewide coverage across Gujarat districts and municipal corporations with GIS coordinates
Deployment Environment	On-Premise / State Data Centre (SDC) / Private Cloud / Hybrid architecture

3. MAJOR ARCHITECTURE COMPONENTS & INGESTION FLOW

Camera Infrastructure & Core Ingestion Flow

Government Cameras / Existing VMS

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Camera Integration & Middleware Layer

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NGINX / API Gateway

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Node.js Backend Services

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Core Data & Messaging Layer

PostgreSQL + PostGIS	Redis (Cache)
Apache Kafka (Events)	OpenSearch (Search)

↓

AI Model API (External Adapter)

↓

AI Detection / Intelligence

Frontend Presentation Flow

Authorized Users
Officers · Operators · Investigators · Admins

↓ *Secure HTTPS*

React + Next.js Web App

↓ *Reverse Proxy / TLS*

NGINX / API Gateway

↓

Node.js Backend Services

Architectural Boundary:

- Frontend connects *exclusively* through NGINX to the Node.js backend.
- AI is an independent, decoupled component invoked via AI Model API.
- Existing VMS/cameras remain untouched; connected via middleware.

4. HIGH-LEVEL ARCHITECTURE DIAGRAM (6-LAYER MODEL)

1. USERS LAYER

- Government Officers
- Operators
- Investigators
- Administrators

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2. PRESENTATION LAYER

React / Next.js Web Application (Desktop-First Control Room Portal)

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3. ACCESS / GATEWAY LAYER

NGINX · Reverse Proxy · TLS Termination · Request Routing

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4. APPLICATION / MIDDLEWARE LAYER (NODE.JS)

Modular REST Application

- Camera Management
- User & Access Mgmt
- GIS / Map
- Live Monitoring
- Search
- Alerts
- Investigations
- Reports

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5. DATA & EVENT LAYER

- **PostgreSQL + PostGIS** — Relational & Geospatial
- **Redis** — Fast State & Cache
- **Apache Kafka** — Event Backbone
- **OpenSearch** — Detection Indexing

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6. VIDEO / AI LAYER — INDEPENDENT PROCESSING & VIEWING PATHS

Existing Government Cameras (RTSP / HTTPS / Supported Streams)

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Camera Integration & Middleware Layer

PATH A: LIVE VIEWING PATH

Live Viewing Stream
↓
WebRTC / HLS
↓
Authorized Users

Raw camera viewing operates continuously even if AI service is offline.

PATH B: AI PROCESSING PATH

AI Processing
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AI Model API (Independent)
↓
AI Detections (JSON Metadata)
↓
Kafka
↓
Search / Alerts / Intelligence

Key Architectural Isolation: AI is an independent, modular microservice connected solely through the **AI Model API**. Live WebRTC/HLS camera viewing and automated AI detection run on distinct, isolated execution paths, ensuring high availability of core control-room viewing.

5. ARCHITECTURAL APPROACH: MODEL 3 – VMS FEDERATION & MIDDLEWARE LAYER

The platform adopts **"Model 3 – VMS Federation and Middleware Layer"** to unify disparate government cameras and existing VMS installations without hardware replacement. It incorporates advanced hybrid concepts for scalable public-sector operations:

Concept	Functional Architectural Rationale
Selective AI Processing	Video feeds undergo AI inference dynamically or on alert triggers rather than continuous indiscriminate processing, optimizing GPU compute.
Dynamic AI Resource Allocation	Workloads are assigned on demand based on priority watchlists, incident triggers, and active departmental requests.
Event-Driven Processing	Apache Kafka decouples high-throughput video detection events from downstream alerting, search indexing, and audit logging.
Centralized Cross-Dept Access	Federated single sign-on with strict multi-department and city-scoped authorization across GSRTC, Police, RTO, and Hospitals.
GIS-Based Camera Management	PostGIS geospatial indexing powers interactive map filtering, bounding-box clustering, and geo-fenced alert boundaries.

6. KEY ARCHITECTURAL PRINCIPLES

- **Integrate existing government camera infrastructure:** Preserve capital investments in cameras, NVRs, and local VMS setups.
- **Centralized access and intelligence layer:** Single authenticated portal for real-time monitoring and cross-department collaboration.
- **AI remains independently deployable:** AI services reside behind API adapters and can be scaled, updated, or restarted independently.
- **Selective live streaming:** On-demand WebRTC streaming prevents network congestion and enforces session-based view limits.
- **Event-driven AI processing:** High-velocity telemetry and detections flow through Kafka topics for asynchronous processing.
- **Horizontally scalable architecture:** Stateless Node.js APIs and clustered data services accommodate future statewide expansion.
- **Secure department and city-based access:** Server-side authorization evaluates user role, department ID, and assigned city jurisdictions.

7. TECHNOLOGY STACK

Component Layer	Selected Technology	Deployment Role & High-Level Specification
Frontend	React, Next.js	Administrative control-room UI, GIS camera clustering, live multi-grid matrix, alert consoles
Backend	Node.js	Application controllers, services, authorization middleware, camera registry, and reporting
Database	PostgreSQL, PostGIS	Relational storage for cameras, users, audit trails, alongside geospatial camera coordinates
Cache	Redis	High-speed caching for active sessions, rate limiting, and real-time operational state
Messaging	Apache Kafka	Distributed streaming backbone for AI detection events, alert topics, and search ingestion
Search	OpenSearch	Full-text and metadata search engine for historical detection events and audit inquiries
Gateway	NGINX	High-performance reverse proxy, TLS termination, load balancing, and request routing
AI Engine	Custom-Trained AI Model	Connected through external AI Model API (vehicle, person, plate, and object analytics)
Streaming Protocols	RTSP, WebRTC/WHEP, HLS, HTTPS	Standardized camera feed ingestion (RTSP/HTTPS) and low-latency browser playback (WebRTC)
Deployment	On-Premise / Private Cloud / Hybrid	Docker-ready containerization suitable for Gujarat State Data Centre (SDC) hosting

Document Evaluation Notice: This High-Level Design (HLD) document is prepared exclusively for technical evaluation of the Gujarat Government Centralized Video Intelligence Platform. All architectural boundaries respect public-sector security protocols and statutory data sovereignty guidelines.